

DESIGN AND ANALYSIS OF ALGORITHMS

Name: Haresh Kumar N L

Reg No: 192425009

Experiment 15: Closest Pair of Points

Code:

```
import math
points = [(1, 2), (4, 5), (7, 8), (3, 1)]
min_distance = float('inf')
closest_pair = None
for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]
        distance = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)
        if distance < min_distance:
            min_distance = distance
            closest_pair = (points[i], points[j])
print("Closest Pair:", closest_pair)
print("Distance =", min_distance)
```

Output:

```
===== RESTART: C:/Users/hk
Closest Pair: ((1, 2), (3, 1))
Distance = 2.23606797749979
```